

Roll No.

TCS-791

B. TECH. (CSE-IS/DS/ML/SE) (SEVENTH SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2022

INTERNET/NETWORK SECURITY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks

1. (a) Explain the following components of Internet architecture : (CO1)
Router, Switch and Access point.

OR

- (b) Demonstrate the working of Internet with the help of an example. (CO1)
2. (a) How would you compare “public IP address and private IP address” and
“static IP address and dynamic IP address” ? (CO2)

OR

- (b) Demonstrate the working of the following DDoS attacks : (CO2)
Volumetric attacks and Application-Layer attacks.
3. (a) Describe the following types of DDoS attacks : (CO3)
ICMP flood and SYN flood

P. T. O.

(2)

OR

(b) How would you compare Network based Intrusion Detection System (NIDS), Host based Intrusion Detection System (HIDS) and Hybrid Intrusion Detection System ? (CO3)

4. (a) Examine the following methods of intrusion detection : (CO4)
Anomaly based detection and Specification based detection.

OR

(b) What are the advantages of Network based Intrusion Detection Systems ? Also discuss few issues associated with the NIDS. (CO4)

5. (a) Why do we use Data Encryption Standard (DES) algorithm ? Provide the details of a single round of the DES algorithm using a figure. (CO5)

OR

(b) Provide the details of the following NIDS performance parameters :
Detection rate and Accuracy. (CO5)